

- 9 A metal block is suspended by a spring balance and is fully submerged in a liquid. When the liquid is replaced with a less dense fluid, the reading on the spring balance
- A increases because upthrust decreases.
 - B increases because the object displaces less fluid.
 - C remains the same because the volume of the block is unchanged.
 - D decreases because of the object changes

Ans: A

Less dense fluid means less upthrust on the metal block. Weight of object remains unchanged. At equilibrium, weight of object is equal to sum of upthrust and tension in spring balance. Smaller upthrust means larger tension in the spring balance.